

5. System Response and LTI Properties

Objective

This experiment is divided into four simulations:

1. Impulse Response of the System
2. Step Response of the System
3. Verification of Linearity Property of LTI Systems
4. Analysis of Discrete-Time System

Experiment 5.1

Impulse Response of the System

Program:

```
// Test Case: Impulse Response of a Continuous-Time System
// Define time range and sampling interval

t_min = -1; // Start time
t_max = 5; // End time
dt = 0.01; // Time step
t = t_min:dt:t_max; // Time vector

// Define the impulse function
x_impulse = zeros(1, length(t));
x_impulse(t == 0) = 1; // Approximate impulse function

// Initialize the output signal with zeros
y_impulse = zeros(1, length(t));

// Define the system parameters
a = 2; // Coefficient for y(t)
b = 1; // Coefficient for x(t)

// Compute the impulse response using the Laplace Transform approach
// Analytical solution:  $y(t) = (b/a) * \exp(-a*t)$  for  $t \geq 0$ 
for i = 1:length(t)
    if t(i) >= 0
        y_impulse(i) = (b / a) * exp(-a * t(i));
    end
end
```

```
// Plot the input and output signals for impulse response

clf; // Clear the current figure

subplot(2,1,1);

plot(t, x_impulse, 'b');

xlabel('t (Time)');

ylabel('x(t)');

title('Input Signal x(t) - Impulse Function');

subplot(2,1,2);

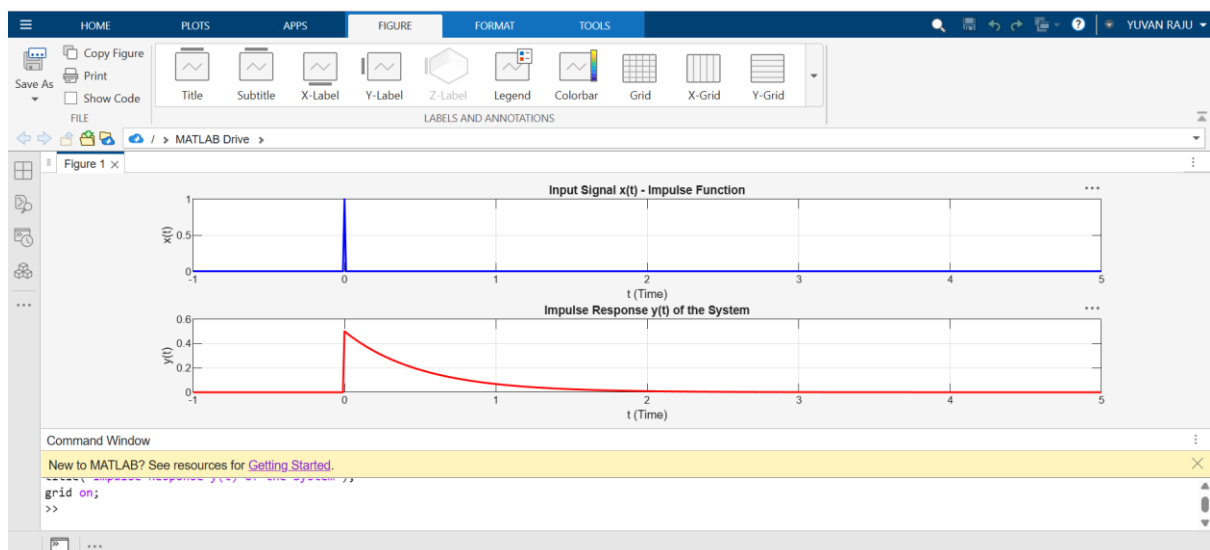
plot(t, y_impulse, 'r');

xlabel('t (Time)');

ylabel('y(t)');

title('Impulse Response y(t) of the System');
```

OUTPUT



Experiment 5.2

Step Response of the System

Program:

```
// Test Case: Step Response of a Discrete-Time System
```

```
// Define the length of the signal
```

```

N = 20; // Length of the signal
// Define the input signal (step function)
u_n = ones(1, N); // Step function

// Define the system parameters
b = [1]; // Numerator coefficients of the system (input)
a = [1, -0.5]; // Denominator coefficients of the system (output)
// Compute the system's response to the step input
y_n = filter(b, a, u_n); // System output for step input
// Define the time index for the signals
n = 0:N-1; // Time index
// Plot the step input and system response
clf; // Clear the current figure
// Plot step input
subplot(2,1,1);
bar(n, u_n, 'b');
xlabel('n');
ylabel('u[n]');
title('Step Input Signal');
// Plot system response to step input
subplot(2,1,2);
bar(n, y_n, 'r');
xlabel('n');
ylabel('y[n]');
title('Step Response of the System');

```

OUTPUT

